

Expt 5 Design of High Pass Filter using Windowing Techniques

5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

Objective:

1. To design an FIR High-Pass Filter using the Rectangular Window technique for the given cutoff frequency, sampling frequency, and filter order specifications.
2. To generate the FIR filter coefficients using the Rectangular Window method in MATLAB and implement the designed filter.
3. To obtain and analyze the magnitude response and phase response of the FIR High-Pass Filter designed using the Rectangular Window.

Procedure:

1. Open MATLAB and create a new script file.
2. Clear the workspace, command window, and close all figure windows.
3. Specify the sampling frequency, cutoff frequency, and filter order.
4. Calculate the normalized cutoff frequency.
5. Design the FIR High-Pass Filter using the `fir1()` function with the 'high' option and a Rectangular Window (`rectwin`).
6. Generate and display the FIR filter coefficients.
7. Obtain the magnitude response and phase response of the designed filter using the `freqz()` function.
8. Plot and analyze the frequency response characteristics of the filter.
9. Verify the high-pass filtering behavior by observing the attenuation of low-frequency components and the passage of high-frequency components.
10. Record the observations and conclude the performance of the FIR High-Pass Filter designed using the Rectangular Window technique.

Program (Matlab Code):

```
clc;  
  
clear;  
  
close all;
```

```

%% --- Input Parameters ---

fp = 1000;      % High-pass cutoff frequency (Hz)
fs = 8000;      % Sampling frequency (Hz)
N = 50;         % Filter order

wc = fp/(fs/2); % Normalized cutoff frequency

%% --- FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));

%% --- Display Coefficients ---

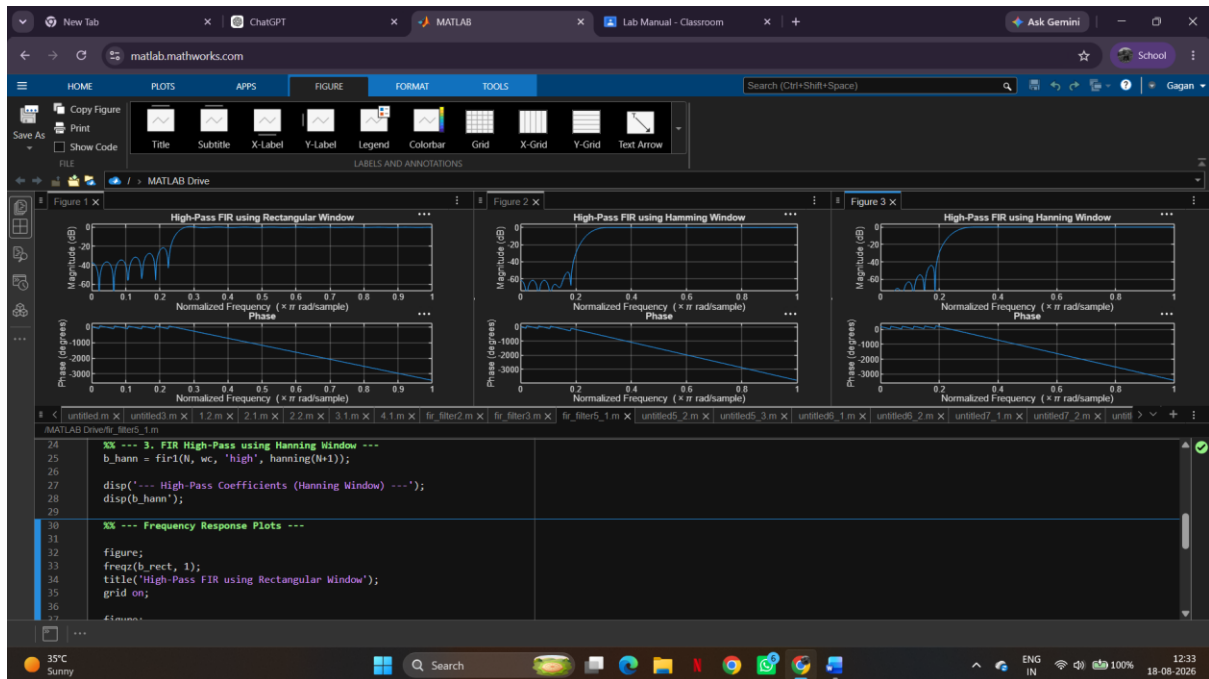
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- Frequency Response Plot ---

figure;
freqz(b_hamm, 1);
title('High-pass FIR using Hamming Window');
grid on;

```

OUTPUT:



--- High-Pass Coefficients (Rectangular Window) ---

-0.0089

-0.0000

0.0097

0.0143

0.0106

-0.0000

-0.0117

-0.0175

-0.0131

-0.0000

0.0148

0.0225

0.0171

-0.0000

-0.0202

-0.0314

-0.0247

-0.0000

0.0318

0.0524

0.0445

-0.0000

-0.0741

-0.1572

-0.2223

0.7407

-0.2223

-0.1572

-0.0741

-0.0000

0.0445

0.0524

0.0318

-0.0000

-0.0247

-0.0314

-0.0202

-0.0000

0.0171

0.0225

0.0148

-0.0000

-0.0131

-0.0175

-0.0117

-0.0000

0.0106

0.0143

0.0097

-0.0000

-0.0089

Result:

1. The FIR High-Pass Filter was successfully designed using the Rectangular Window technique for the specified cutoff frequency, sampling frequency, and filter order.
2. The FIR filter coefficients were successfully generated using MATLAB and the designed filter was implemented.
3. The magnitude response and phase response of the designed FIR High-Pass Filter were obtained and analyzed. The magnitude response showed attenuation of low-frequency components and passage of high-frequency components, confirming the desired high-pass filtering behavior. The phase response exhibited the expected linear-phase characteristics of an FIR filter.